

AP1_Assignment 3_22167449_NGUYEN Ngoc Nhu Y.pdf *by Ngoc Nhu Y Nguyen*

Submission date: 22-Nov-2025 02:27AM (UTC+1100)

Submission ID: 2823144603

File name: AP1_Assignment_3_22167449_NGUYEN_Ngoc_Nhu_Y.pdf (548.86K)

Word count: 2496

Character count: 13078

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ASSIGNMENT COVER SHEET

STUDENT DETAILS

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UNIT AND TUTORIAL DETAILS

Unit name: Analytics Programming Unit number: AP-T325WSD-1

Tutorial Group: _____ Class day and time: 15:30 Sundat

Lecturer or Tutor name: Assoc. Prof. NGUYEN Tan Luy

ASSIGNMENT DETAILS

Title: Assignment 3: T2 2025

Length: 14 pages Due date: 21/11/2025 Date submitted: 21/11/2025

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ANALYTICS PROGRAMMING

Assignment 3

Name: Nguyen Ngoc Nhu Y

Western Sydney University

Class: AP-T325WSD-1

Lecturer: Assoc. Prof. NGUYEN Tan Luy

Repository Link:

https://github.com/nhuwys/AP1_Assignment-3_22167449_NGUYEN-Ngoc-Nhu-Y

Assignment 3

NGUYEN Ngoc Nhu Y

Task 1

```
#Read the dataset and replace "?" with NA
engine <- read.csv("Engine.csv", na.strings = "?")
automobile <- read.csv("Automobile.csv", na.strings = "?")
maintenance <- read.csv("Maintenance.csv", na.strings = "?")
```

```
#Inspect the structure of the dataset "Engine.csv"
str(engine)
```

```
'data.frame':  88 obs. of  8 variables:
 $ EngineModel : chr  "E-0001" "E-0002" "E-0003" "E-0004" ...
 $ EngineType  : chr  "dohc" "ohcv" "ohc" "ohc" ...
 $ NumCylinders: chr  "four" "six" "four" "five" ...
 $ EngineSize  : int  130 152 109 136 136 131 131 108 164 164 ...
 $ FuelSystem  : chr  "mpfi" "mpfi" "mpfi" "mpfi" ...
 $ Horsepower  : int  111 154 102 115 110 140 160 101 121 121 ...
 $ FuelTypes   : chr  "gas" "gas" "gas" "gas" ...
 $ Aspiration  : chr  "std" "std" "std" "std" ...
```

```
#Inspect the structure of the dataset "Automobile.csv"
str(automobile)
```

```
'data.frame':  204 obs. of  13 variables:
 $ PlateNumber : chr  "53N-001" "53N-002" "53N-003" "53N-004" ...
 $ Manufactures : chr  "Alfa-romero" "Alfa-romero" "Audi" "Audi" ...
 $ BodyStyles   : chr  "convertible" "hatchback" "sedan" "sedan" ...
 $ DriveWheels  : chr  "rwd" "rwd" "fwd" "4wd" ...
 $ EngineLocation: chr  "front" "front" "front" "front" ...
 $ WheelBase    : num  88.6 94.5 99.8 99.4 99.8 ...
 $ Length       : num  169 171 177 177 177 ...
 $ Width        : num  64.1 65.5 66.2 66.4 66.3 71.4 71.4 71.4 67.9 64.8 ...
 $ Height       : num  48.8 52.4 54.3 54.3 53.1 55.7 55.7 55.9 52 54.3 ...
 $ CurbWeight   : int  2548 2823 2337 2824 2507 2844 2954 3086 3053 2395 ...
 $ EngineModel  : chr  "E-0001" "E-0002" "E-0003" "E-0004" ...
 $ CityMpg      : int  21 19 24 18 19 19 19 17 16 23 ...
 $ HighwayMpg   : int  27 26 30 22 25 25 25 20 22 29 ...
```

```
#Inspect the structure of the dataset "Maintenance.csv"
str(maintenance)
```

```

6
'data.frame': 374 obs. of 7 variables:
 $ ID      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ PlateNumber: chr  "53N-001" "53N-001" "53N-001" "53N-001" ...
 $ Date      : chr  "15/02/2024" "16/03/2024" "15/04/2024" "15/05/2024" ...
 $ Troubles   : chr  "Break system" "Transmission" "Suspected clutch" "Ignition (finding)" ...
 $ ErrorCodes : int  -1 -1 -1 1 -1 1 1 0 -1 -1 ...
 $ Price      : int  110 175 175 180 85 1000 180 0 180 180 ...
 $ Methods    : chr  "Replacement" "Replacement" "Adjustment" "Adjustment" ...

```

```

#Count total missing values after replacing "?" with NA
cat("Missing values in Engine:", sum(is.na(engine)), "\n")

```

Missing values in Engine: 6

```

cat("Missing values in Automobile:", sum(is.na(automobile)), "\n")

```

Missing values in Automobile: 0

```

cat("Missing values in Maintenance:", sum(is.na(maintenance)), "\n")

```

Missing values in Maintenance: 0

```

# Check which columns contain NA in the Engine dataset
colSums(is.na(engine))

```

EngineModel	EngineType	NumCylinders	EngineSize	FuelSystem	Horsepower
0	5	0	0	0	1
FuelTypes	Aspiration				
0	0				

```

# Count number of rows in Engine affected by missing values
rows_with_na <- sum(rowSums(is.na(engine)) > 0)
cat("Rows with missing values in Engine:", rows_with_na, "\n")

```

Rows with missing values in Engine: 6

Notes:

Only Engine has missing values “?” (Engine = 6, Automobile & Maintenance = 0).

-> only Engine has affected rows (Missing values occur only in Horsepower)

-> only Engine needs cleaning

When inspecting the column, it is appeared that there are:

- 5 missing entries occur in EngineType
- 1 missing entry occurs in Horsepower

Replacing “?” with NA does not change the distribution because no actual values are modified; NA only marks missing entries.

```

#Convert categorical variables to factors
automobile$BodyStyles <- as.factor(automobile$BodyStyles)
engine$FuelTypes <- as.factor(engine$FuelTypes)
maintenance$ErrorCodes <- as.factor(maintenance$ErrorCodes)

#Compute the median Horsepower for imputation
median_hp <- median(engine$Horsepower, na.rm = TRUE)

#Replace the missing Horsepower values with median
engine$Horsepower[is.na(engine$Horsepower)] <- median_hp

```

Horsepower is a numerical variable

-> use median -> resistant to outliers and preserves the central tendency

Note on Engine Type missing values: Engine Type is categorical; missing entries were retained as NA and it is only required to replace missing Horsepower values.

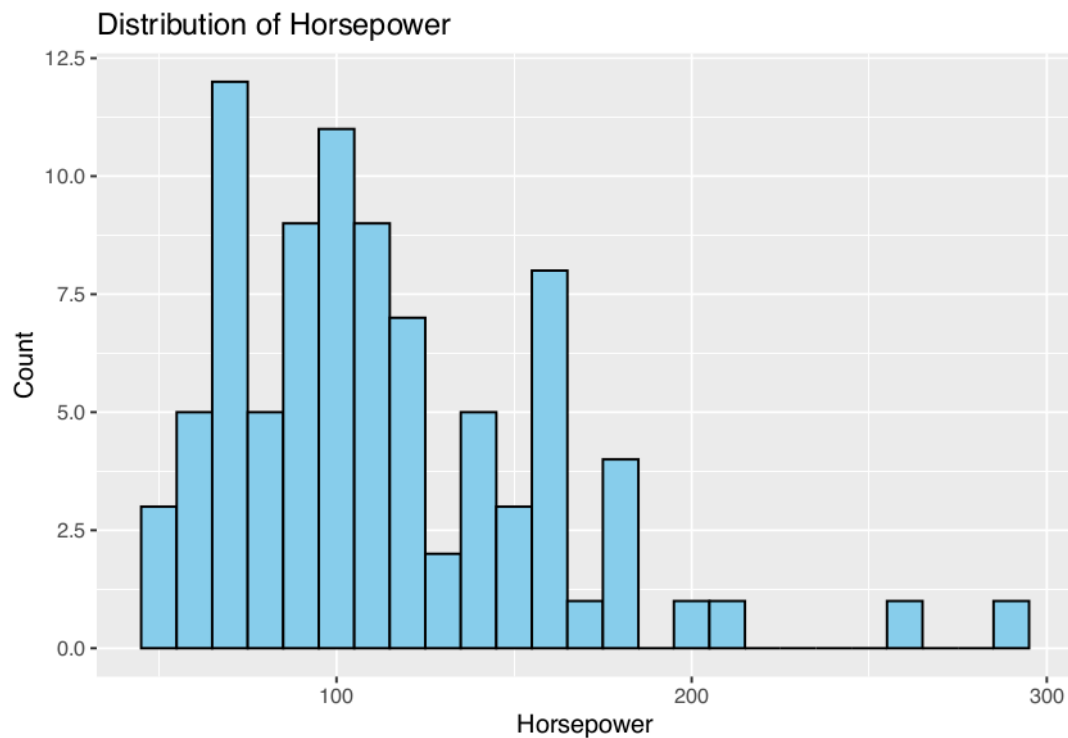
Distribution of Horsepower Across Engine Types

```

# Distribution of Horsepower Across Engine Types
library(ggplot2)

#Visualize Horsepower distribution
ggplot(engine, aes(x = Horsepower)) +
  geom_histogram(binwidth = 10, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Horsepower",
       x = "Horsepower",
       y = "Count")

```



The histogram indicates a strongly right-skewed distribution. The majority of Horsepower values are concentrated between 50 and 170, suggesting that most engines in the dataset belong to small to mid-sized vehicles. However, the long right tail indicates several high-horsepower engines, which act as outliers.

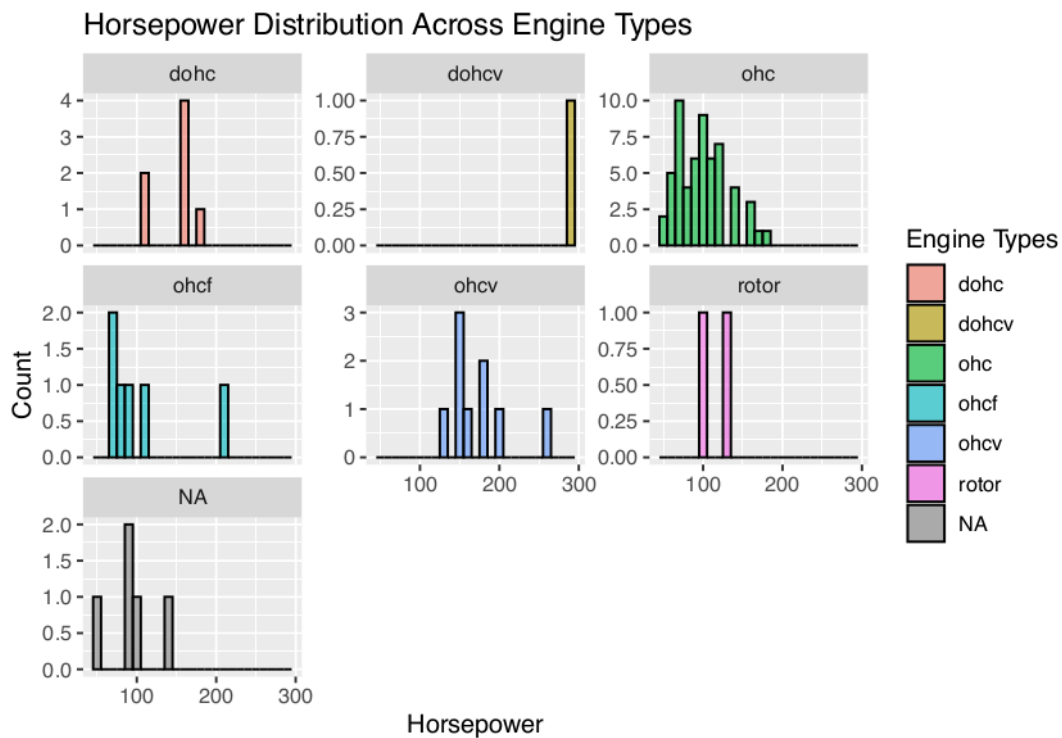
Since the missing values were replaced with the median horsepower, the distribution remains stable and does not shift. This is expected because the median is robust to extreme values and represents the central tendency of the dataset.

Task 2

Horsepower Distribution Across Engine Types

```
# Distribution of Horsepower Across Engine Types
library(ggplot2)

# Histogram of Horsepower for each engine type
ggplot(engine, aes(x = Horsepower, fill = EngineType)) +
  geom_histogram(binwidth = 10, alpha = 0.6, color = "black") +
  facet_wrap(~EngineType, scales="free_y") +
  labs(title = "Horsepower Distribution Across Engine Types",
       x = "Horsepower",
       y = "Count",
       fill = "Engine Types")
```



The distribution of horsepower varies noticeably across different engine types. OHC engines appear most frequently in the dataset and generally fall within the low- to mid-range horsepower levels, indicating that they are typically used in standard consumer vehicles. In contrast, DOHC and OHCV engines display a wider spread and extend into higher horsepower ranges, suggesting that these engine types are associated with stronger performance capabilities. Rotor engines are rare and exhibit a very narrow horsepower distribution, highlighting their limited use in this dataset.

=> The engine type is closely related to the achievable horsepower, with more advanced configurations tending to produce greater power.

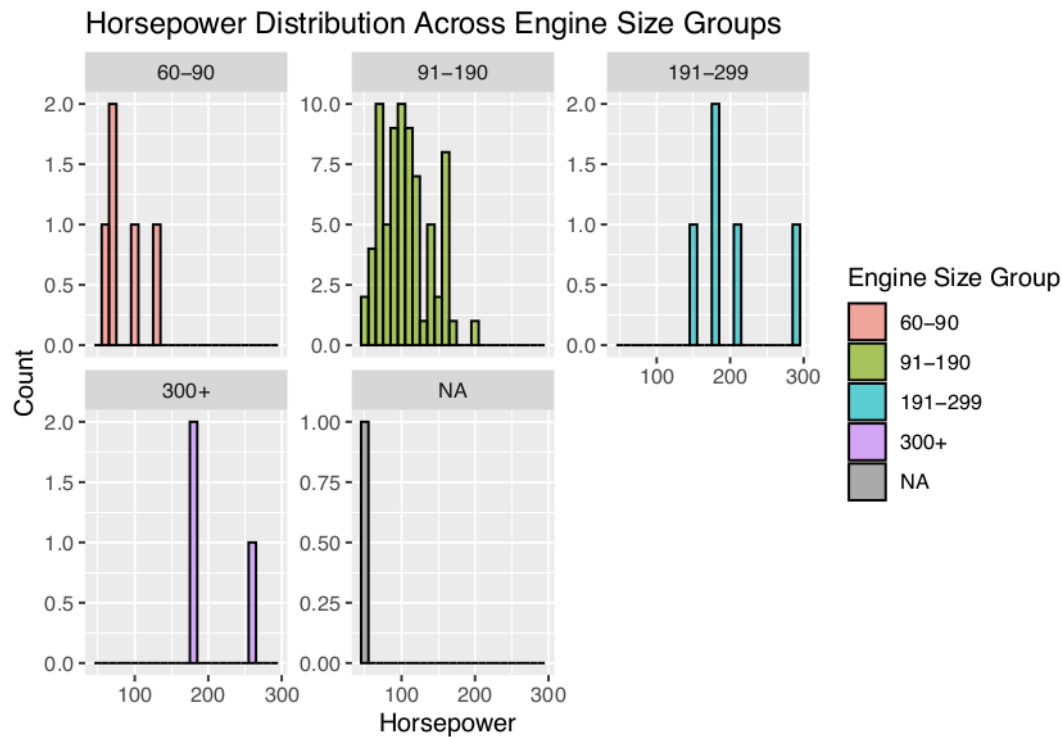
Horsepower Distribution Across Engine Size Groups

```
# Create Engine Size Groups
engine$SizeGroup <- cut(engine$EngineSize,
  breaks = c(60, 90, 190, 299, Inf),
  labels = c("60-90", "91-190", "191-299", "300+"))

#Create table for Engine Size Groups
table (engine$SizeGroup)
```

60-90	91-190	191-299	300+
5	74	5	3


```
#Facet_wrap Histogram of Horsepower by Size Groups
ggplot(engine, aes(x = Horsepower, fill = SizeGroup)) +
  geom_histogram(binwidth = 10, alpha = 0.6, color = "black") +
  facet_wrap(~ SizeGroup, scales = "free_y") +
  labs(title = "Horsepower Distribution Across Engine Size Groups",
       x = "Horsepower",
       y = "Count",
       fill = "Engine Size Group")
```



When horsepower is examined across the engine size groups, the distribution reveals that the majority of engines belong to the 91-190 size range, which corresponds to mid-sized engines commonly found in everyday vehicles. These engines have a broad horsepower spread but remain mostly within standard performance levels. Smaller engines in the 60-90 group show notably lower horsepower values, reflecting their limited power output. Engines in the 191-299 group shift toward higher horsepower ranges but occur much less frequently, while engines in the 300+ group are extremely rare and represent high-performance or specialized models.

=> Clear relationship between engine size and horsepower, where larger engines tend to support higher power capacity.

Task 3

3.1 Do diesel cars have higher average CityMpg than gasoline cars?

```
t_test_fuel <- t.test(CityMpg ~ FuelTypes,  
                      data = merge(automobile, engine, by = "EngineModel"))  
t_test_fuel
```

Welch Two Sample t-test

```
data: CityMpg by FuelTypes  
t = 3.9004, df = 22.592, p-value = 0.0007392  
alternative hypothesis: true difference in means between group diesel and group gas is not equal to 0  
95 percent confidence interval:  
 2.824648 9.218138  
sample estimates:  
mean in group diesel    mean in group gas  
      30.30000         24.27861
```

To determine whether diesel vehicles have higher average CityMpg than gasoline vehicles:

-> The results show a statistically significant difference between the two groups with:

- $t = 3.90$
- $df = 22.6$
- $p = 0.00074$

-> diesel: 30.30 MPG, gasoline: 24.28 MPG

-> Diesel cars have significantly higher CityMpg than gasoline cars

-> The 95% confidence interval (2.82 to 9.22 MPG) confirms that this difference is meaningful

-> Since diesel engines generally operate more efficiently at lower speeds and produce better fuel economy in city driving

Note: check whether two groups have different averages -> t-test: best choice

3.2 How does DriveWheels affect fuel efficiency (CityMpg and HighwayMpg)?

```
# Mean CityMpg by DriveWheels  
city_means <- aggregate(CityMpg ~ DriveWheels, data = automobile, mean)  
city_means
```

```
DriveWheels CityMpg  
1          4wd 23.11111  
2          fwd 28.32500  
3          rwd 20.53333
```

```
# One-way ANOVA for CityMpg
anova_city <- aov(CityMpg ~ DriveWheels, data = automobile)
summary (anova_city)

          Df Sum Sq Mean Sq F value Pr(>F)
DriveWheels  2   2844  1422.1   47.64 <2e-16 ***
Residuals 201   6000    29.9
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Mean HighwayMpg by DriveWheels
highway_means <- aggregate(HighwayMpg ~ DriveWheels, data = automobile, mean)
highway_means
```

```
DriveWheels HighwayMpg
1          4wd   27.22222
2          fwd   34.23333
3          rwd   25.64000
```

```
# One-way ANOVA for HighwayMpg
anova_highway <- aov(HighwayMpg ~ DriveWheels, data = automobile)
summary (anova_highway)
```

```
          Df Sum Sq Mean Sq F value Pr(>F)
DriveWheels  2   3526  1763.2   56.77 <2e-16 ***
Residuals 201   6242    31.1
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

To **examine** whether DriveWheels affects fuel efficiency:

DriveWheels has a significant effect on both CityMpg and HighwayMpg
(p-value < 2e-16 < 0.001)

Besides,

- FWD cars use the **least** fuel, giving them the highest MPG
- 4WD cars use the **most** fuel, resulting in the lowest MPG
- RWD vehicles fall between the two

Note: DriveWheels strongly influences fuel economy, and ANOVA is the correct test for comparing three or more groups

3.3 Filtering engines with trouble and Identifying Top 5 Trouble Types

```
# Filter trouble-related cases
trouble <- maintenance[maintenance$ErrorCodes != 0, ]

# List top 5 most common troubles
top5_troubles <- sort(table(trouble$Troubles), decreasing = TRUE)[1:5]
top5_troubles
```

Cylinders	38	Chassis Ignition (finding)	25	Noise (finding)	22	19
Worn tires	16					

Cylinders are the most common troubles (38)

Then Chasis (25), Ignition Issues (22), Noise-related Issues (19), and Worn tires (16)

Note: Count how many times each category appears -> frequency table

3.4 Association Between Engine Type and Trouble Type

```
trouble$HasTrouble <- ifelse(trouble$Troubles == "none" | is.na(trouble$Troubles), 0, 1)
table_simple <- table(trouble$ErrorCodes, trouble$HasTrouble)
chisq.test(table_simple)
```

5 Chi-squared test for given probabilities

```
data: table_simple
X-squared = 174.4, df = 2, p-value < 2.2e-16
```

- Chi-square test -> there is significant association (p-value < 2.2e-16 < 0.001)
- Vehicles with ErrorCodes # 0 -> more likely to have trouble
- ErrorCodes is a useful indicator of suspected issues

Note: two categorical variables are connected -> Chi-square test

Task 4

4.1 Which error type (ErrorCodes) occurs most frequently

```
# Count frequency of each error type
error_counts <- table(maintenance$ErrorCodes)
error_counts
```

```
-1    0    1
164  28 182
```

- ErrorCode = 1 (confirmed trouble): most frequent
- ErrorCode = -1 (suspected trouble): second
- ErrorCode = 0 (no trouble): least frequent

4.2 Analyze the factors that might influence the Maintenance Methods

```
# Merge datasets and filter trouble cases
library(dplyr)
```

```
trouble_data <- merge(automobile, engine, by = "EngineModel") %>%
  merge(maintenance, by = "PlateNumber") %>%
  filter(ErrorCodes != 0)
```

```
# FuelTypes vs Methods
```

```
table_Fuel_Methods <- table(trouble_data$FuelTypes, trouble_data$Methods)
table_Fuel_Methods
```

	Adjustment	Replacement	Urgent care
diesel	7	21	0
gas	128	177	29

```
# Chi-squared test
```

```
chisq.test(table_Fuel_Methods)
```

Pearson's Chi-squared test

```
data: table_Fuel_Methods
X-squared = 5.9481, df = 2, p-value = 0.0511
```

```
#BodyStyles vs Methods
```

```
table_Body_Methods <- table(trouble_data$BodyStyles, trouble_data$Methods)
table_Body_Methods
```

	Adjustment	Replacement	Urgent care
convertible	7	8	0
hardtop	2	4	2
hatchback	48	68	9
sedan	63	94	15
wagon	15	24	3

```
#Chi-squared test
```

```
chisq.test(table_Body_Methods)
```

Pearson's Chi-squared test

```
data: table_Body_Methods
X-squared = 5.1868, df = 8, p-value = 0.7374
```

```
#Troubles vs Methods
```

```
table_Troubles_Methods <- table(trouble_data$Troubles, trouble_data$Methods)
table_Troubles_Methods
```

	Adjustment	Replacement	Urgent care
Air conditioner	9	0	0
Bearing	0	3	0
Brake fluid	0	0	13
Break system	0	10	0
Cam shaft	0	12	0
Chassis	0	25	0
Crank shaft	0	7	0
Cylinders	0	39	0
ECU's power	8	0	0
Fans	0	13	0
Front axle	0	3	0
Gear box (finding)	0	4	0
Ignition	9	0	0
Ignition (finding)	23	0	0
Loss of driving ability	0	0	16
Noise (finding)	20	0	0
O2 sensors	0	1	0
Oil filter	0	4	0
Painting	0	15	0
Pedals	7	0	0
Pressure sensors	0	11	0
Real axle	0	9	0
Side slip	15	0	0
Steering wheel	0	9	0
Stroke	0	4	0
Suspected battery	10	0	0
Suspected clutch	13	0	0
Suspension	6	0	0
Temperature sensors	0	8	0
Transmission	0	5	0
Valve clearance	15	0	0
Worn tires	0	16	0

```
#Chi-squared test
```

```
chisq.test(table_Troubles_Methods)
```

Pearson's Chi-squared test

```
data: table_Troubles_Methods
```

```
X-squared = 724, df = 62, p-value < 2.2e-16
```

I chose Troubles, FuelTypes, and BodyStyles to clearly examine how different factors may influence the Maintenance Methods applied to trouble vehicles

Troubles x Methods (Significant)

- Chi-square p-value $< 2.2\text{e-}16 < 0.05$ (strongest significant relationship)
- Clear pattern:
 - Severe issues (Cylinders, Chassis, Cam shaft, Break system) -> mostly Replacement.
 - Minor or diagnostic issues (Noise, Ignition finding, Side slip) -> mostly Adjustment.

=> Type of trouble strongly determines the Maintenance Methods

FuelTypes (Near-significant)

- Chi-square p-value $= 0.0511 > 0.05$ (but closest to significance among all non-significant factors)
- Clear trend:
 - Diesel vehicles -> higher proportion of Replacement.
 - Gasoline vehicles -> more balanced distribution across methods.

=> Fuel type hints at the underlying stress level on the engine, which affects the likelihood of needing Maintenance Methods (Replacement, Adjustment)

FuelTypes (Not significant)

- Chi-square p-value $= 0.7374 > 0.005$ -> no statistical relationship

=> The shape or design of the car body does not influence how the vehicle must be repaired

Task 5:

```
ls
cd '/Users/nguyenngocnhuy/Downloads/A3_22167449_NGUYEN Ngoc Nhu Y'
git init
git add A3.rmd
git commit -m "Task 1:Count total missing values after replacing "?" with NA"
git add A3.rmd
git commit -m "Task 1: Count missing values"
git add A3.rmd
git commit -m "Task 1:check columns"
git add A3.rmd
git commit -m "Task 1:Count number of rows in Engine affected by missing values"
git add A3.rmd
git commit -m "Task 1:Notes"
git add A3.rmd
git commit -m "Task 1:Notes"
git add A3.rmd
git commit -m "Task 1: Convert categorical variables to factors"
git add A3.rmd
git commit -m "Task 1: Compute the median Horsepower for imputation"
git add A3.rmd
git commit -m "Task 1:Replace the missing Horsepower values with median"
git add A3.rmd
git commit -m "Task 1:Notes"
git add A3.rmd
git commit -m "Task 1:Distribution of Horsepower Across Engine Types"
git add A3.rmd
git commit -m "Task 1:Interpretation"
git add A3.rmd
git commit -m "Task 2"
git add A3.rmd
git commit -m "Task 2: Horsepower Distribution Across Engine Types"
git add A3.rmd
git commit -m "Task 2: Interpretation"
git add A3.rmd
git commit -m "Task 2:Create Engine Size Groups"
git add A3.rmd
git commit -m "Task 2: Create Histogram Horsepower Distribution Across Engine Size Groups"
git add A3.rmd
git commit -m "Task 2: Interpretation"
git add A3.rmd
git commit -m "Task 3"
git add A3.rmd
git commit -m "Task 3: 3.1"
git add A3.rmd
git commit -m "Task 3: 3.1 Interpretation and Notes"
git add A3.rmd
git commit -m "Task 3: 3.2"
git add A3.rmd
git commit -m "Task 3: 3.2 One-way ANOVA for CityMpg"
git add A3.rmd
git commit -m "Task 3: 3.2 One-way ANOVA for HighwayMpg"
git add A3.rmd
git commit -m "Task 3: 3.2 Interpretation"
git add A3.rmd
git commit -m "Task 3: 3.3 Filtering engines with trouble and Identifying Top 5 Trouble Types"
```



```
git add A3.rmd
git commit -m "Task 3: 3.2 fixed"
git add A3.rmd
git commit -m "Task 3: 3.3 Interpretation"
git add A3.rmd
git commit -m "Task 3: 3.4 Association Between Engine Type and Trouble Type"
git add A3.rmd
git commit -m "Task 3: 3.4 Interpretation"
git add A3.rmd
git commit -m "Task 4: 4.1 Interpretation"
git add A3.rmd
git commit -m "Task 4: 4.2 Merge datasets and filter trouble cases"
git add A3.rmd
git commit -m "Task 4: 4.2 Choosing Factors: FuelTypes vs Methods"
git add A3.rmd
git commit -m "Task 4: 4.2 Choosing Factors: BodyStyles vs Methods"
git add A3.rmd
git commit -m "Task 4: 4.2 Choosing Factors: Troubles vs Methods"
git add A3.rmd
git commit -m "Task 4: 4.2 Interpretation"
git add A3.pdf
git commit -m "Add final knitted PDF"
git remote add origin https://github.com/nhuwys/AP1_Assignment-3_22167449_NGUYEN-Ngoc-Nhu-Y.git
git branch -M main
git push -u origin main
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